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A Seminar Report

On

Blue Brain Technology

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Submitted By

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CERTIFICATE

Certified that the Technical Seminar Entitled **"Blue Brain Technology"** carried out by **MADHU SUDHAN S**, bearing **1DT19CS074,** bonafide student of Dayananda Sagar Academy of Technology and Management, in partial fulfillment for the award of the **BACHELOR OF ENGINEERING** in Computer Science and Engineering from **Visvesvaraya Technological University, Belagavi** during the year 2022-2023. The report has been approved as it satisfies the academic requirements in respect of the Technical Seminar work prescribed for the said degree.

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Date: **MADHU SUDHAN S [1DT19CS074]**

ABSTRACT

Today scientists are in research to create an artificial brain that can think, respond, take decision, and keep anything in memory. The main aim is to upload human brain into machine. So that man can think, take decision without any effort. After the death of the body, the virtual brain will act as the man. So, even after the death of a person we will not loose the knowledge, intelligence, personalities, feelings and memories of that man, that can be used for the development of the human society. Technology is growing faster than every thing.

IBM is now in research to create a virtual brain, called “Blue brain”. If possible, this would be the first virtual brain of the world. IBM, in partnership with scientists at Switzerland’s Ecole Polytechnique Federale de Lausanne’s (EPFL) Brain and Mind Institute will begin simulating the brain’s biological systems and output the data as a working 3-dimensional model that will recreate the high-speed electro-chemical interactions that take place within the brain’s interior. These include cognitive functions such as language, learning, perception and memory in addition to brain malfunction such as psychiatric disorders like depression and autism. From there, the modeling will expand to other regions of the brain and, if successful, shed light on the relationships between genetic, molecular and cognitive functions of the brain.

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CHAPTER 1

INTRODUCTION

Brain of humans are the most valuable creation of God. A person is called intelligent because of his brain. The brain translates the information delivered by the impulses, which then enables the person to react. But we loss the knowledge of a brain when the body is destroyed after the death of man. That knowledge might have been used for the development of the human society. What happen if we create a brain and up load the contents of natural brain into it?

1.1 What Is Blue Brain?

The Blue Brain Project is assumed to be the first one to explore about a true "Artificial Intelligence" via the process of reverse engineering and also the effort to reverse engineering a human brain. The vision behind Virtual Brain will help shed some light on some aspects of human recognition. The Blue Brain Project aims to build a full computer model of a functioning brain to simulate drug treatments or any other brain related problems. Blue gene supercomputer constructed by IBM was a machine first used by Blue Brain Project and then a term Blue Brain was introduced. It can be implemented by using supercomputer, the fastest type but quite expensive and are assist for special tasks which require abundant amount of mathematical computations, like weather forecasting employs a supercomputer. The back pane of Blue Brain is Artificial Intelligence, a technology which builds intelligent machines and imparts intelligent agents. Knowledge, learning, reasoning, planning, communication and perception are the main goals of its research.

1.2 Why We Need Blue Brain?

Today we are developed because of our intelligence. Intelligence is the inborn quality that can't be created. Some people have this quality, so that they can think up to such an extent where other can't reach. Human society is always need of such intelligence and such an intelligent brain to have with. But the intelligence is lost along with the body after the death. The virtual brain is a solution to it. The brain and intelligence will alive even after the death. We often face difficulties in remembering things such as people's names, their birthdays, and the spellings of words, proper grammar, important dates, history, facts etc… In the busy life every one want to be relaxed. Can't we use any machine to assist for all these? Virtual brain may be the solution to it. What if we upload ourselves into computer, we were simply aware of a computer, or maybe, what if we lived in a computer as a program?

1.3 Motivation

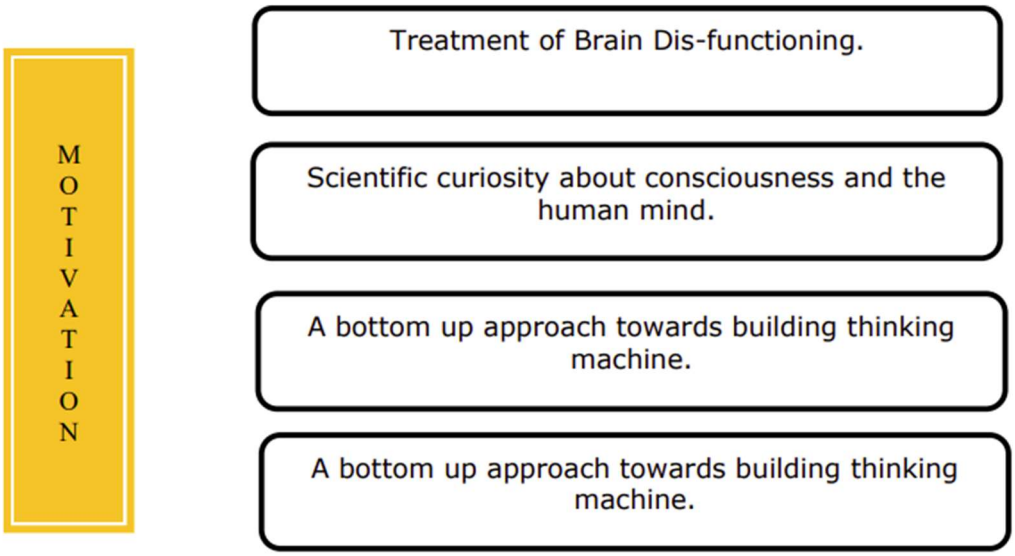


Fig : 1.3 Motivation

A very good example of utilization of blue brain is the case "short term memory". In some movies we might have noticed that a person might be having short term memories.

CHAPTER 2

SYSTEM ARCHITECTURE

2.1 Data Acquisition

We need to study different types of neurons and catalogue them. In involves taking brain slices and placing them under a microscope, measuring the shape and electrical activity of individual neurons. Morphology (A branch of biology dealing with the study of form and structure of organisms and their specific structural features, Comparative morphology, Functional morphology and Experimental morphology are different branches of morphology), Electrophysiology behavior (A minimally invasive procedure which test the electrical conduction system of the heart to assess the electrical activity and conduction pathways of the heart, it enables twelve living neurons to be concurrently patched and their electrical activity recorded), location within the cortex and their population density are the different factors by which neurons are typed. Form, function and positioning of neurons are obtained by translating these observations into mathematical algorithms. By the help of these algorithms, biologically-realistic virtual neurons ready for simulation are generated.

2.2 Simulation

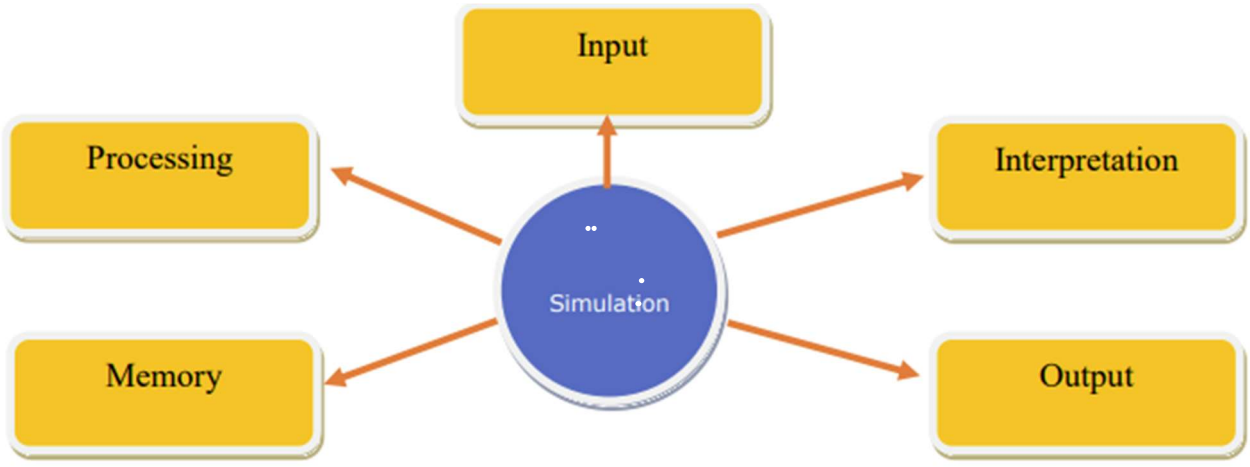


Fig 2.2 Simulation

2.2.1 Input

In the nervous system of our body, neurons are responsible for message passing. Sensory cells convey the input to our body. Electric impulses are produced by sensory cells which are received by the neurons. Further, electric impulses (through the silicon chips of artificial neurons) are transferred to the brain by neurons.

2.2.2 Interpretation

Brain receives the electric impulses from the neurons which are interpreted by means of registers. Different states of brain can be accomplished by the different values in these registers

2.2.3 Output

On the basis of the states of the registers, the electric impulses are sending by the brain representing the responses which are then received by the sensory cells to respond .The sensory cells of which part of our body is going to receive that, it depends on the state of the neurons in the brain at that time.

2.2.4 Memory

We can store certain information (states) permanently in our brain by certain neurons .On the basis of our requirement, the brain interprets those states and thus past things can be gathered. To do so we demand the neurons, to permanently show the clear vision of the Brain's States.

2.2.5 Processing

By the use of some stored states, computation will be performed by the computer. Logical and arithmetic calculations are done in our neural circuitry. To produce the output stored past experiences and current input received are used.

2.3 Visualization of Results

Huge amount of data are generated by running the Blue Brain simulation. Thousands of times analyses of individual neurons must be repeated. Data can be analyzed by using massively parallel computers where it is created (serverside analysis for experimental data, online analysis during simulation).

A visual exploration of the circuit is an important part of the analysis (given the geometric complexity of the column). It is invaluable for an immediate verification of single cell activity by mapping the simulation data onto the morphology. The Blue Gene has been translated into a 3D visual representation of the column to design a visualization interface. A challenging task is the visualization of the neurons given the fact that a series of ten thousand neurons clustered in high quality mesh accounts for essentially 1 billion triangles for which about hundred GB of management data is required. To study in detail using further simulations, visual interface makes it easy to quickly identify those areas and to show electrical activity in the brain, a visual representation could be also used.

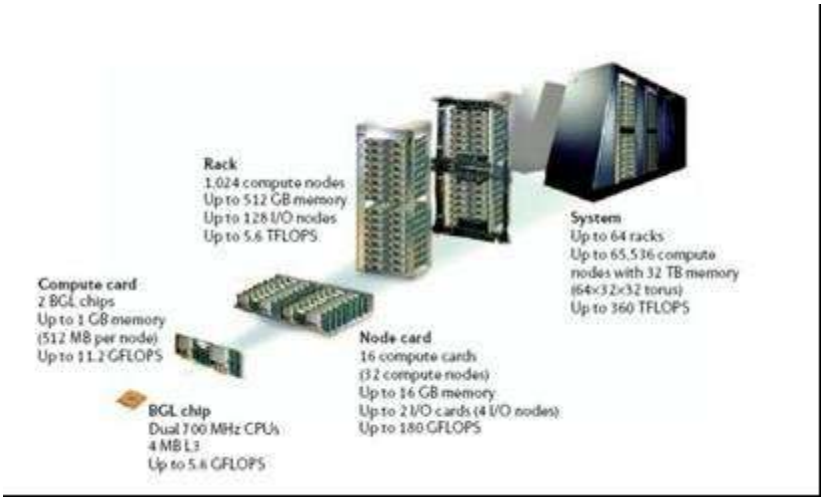


Fig 2.3 The Blue Gene/L supercomputer architecture

CHAPTER 3

REFLECTION NOTES ON IMPLEMENTATION

First, it is helpful to describe the basic manners in which a person may be uploaded into a computer. Raymond Kurzweil recently provided an interesting paper on this topic. In it, he describes both invasive and noninvasive techniques. The most promising is the use of very small robots, or nanobots. These robots will be small enough to travel throughout our circulatory systems. Traveling into the spine and brain, they will be able to monitor the activity and structure of our central nervous system. They will be able to provide an interface with computers that is as close as our mind can be while we still reside in our biological form. Nanobots could also carefully scan the structure of our brain, providing a complete readout of the connections between each neuron. They would also record the current state of the brain. This information, when entered into a computer, could then continue to function as us.



Fig 3.1 Nano Bot

How natural brain works?

The human ability to feel, interpret and even see is controlled, in computer like calculations, by the magical nervous system. Yes, the nervous system is quite like magic because we can't see it, but its working through electric impulses through your body.

One of the worlds most "intricately organized" electron mechanisms is the nervous system. Not even engineers have come close to making circuit boards and computers as delicate and precise as the nervous system. To understand this system, one has to know the three simple functions that it puts into action: sensory input, integration, motor output.

Sensory input:

When our eyes see something or our hands touch a warm surface, the sensory cells, also known as Neurons, send a message straight to your brain. This action of getting information from your surrounding environment is called sensory input because we are putting things in your brain by way of your senses.

Integration:

Integration is best known as the interpretation of things we have felt, tasted, and touched with our sensory cells, also known as neurons, into responses that the body recognizes. This process is all accomplished in the brain where many, many neurons work together to understand the environment.

Motor Output:

Once our brain has interpreted all that we have learned, either by touching, tasting, or using any other sense, then our brain sends a message through neurons to effecter cells, muscle or gland cells, which actually work to perform our requests and act upon our environment.

The word motor output is easily remembered if one should think that our putting something out into the environment through the use of a motor, like a muscle which does the work for our body.

How we see, hear, feel, smell, and take decision:

Nose

Once the smell of food has reached your nose, which is lined with hairs, it travels to an olfactory bulb, a set of sensory nerves. The nerve impulses travel through the olfactory tract, around, in a circular way, the thalamus, and finally to the smell sensory cortex of our brain, located between our eye and ear, where it is interpreted to be understood and memorized by the body.

Eye

Seeing is one of the most pleasing senses of the nervous system. This cherished action primarily conducted by the lens, which magnifies a seen image, vitreous disc, which bends and rotates an
image against the retina, which translates the image and light by a set of cells. The retina is at the back of the eye ball where rods and cones structure along with other cells and tissues covert the image into nerve impulses which are transmitted along the optic nerve to the brain where it is kept for memory.

Tongue

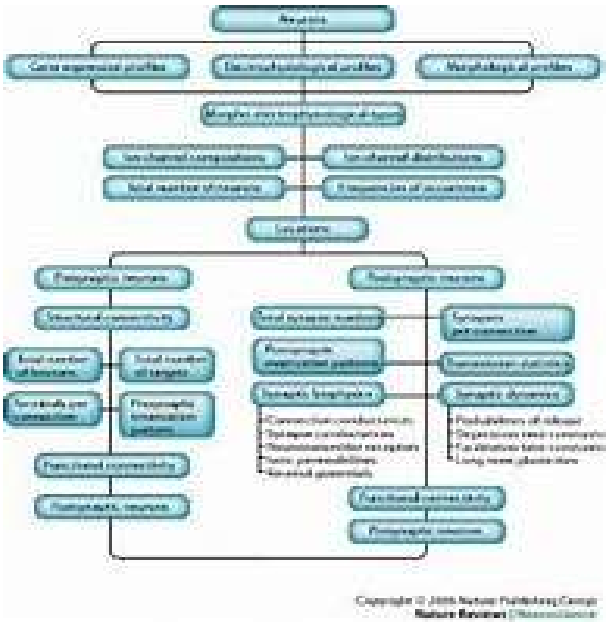
A set of microscopic buds on the tongue divide everything we eat and drink into four kinds of taste: bitter, sour, salty, and sweet. These buds have taste pores, which convert the taste into a nerve impulse and send the impulse to the brain by a sensory nerve fiber. Upon receiving the message, our brain classifies the different kinds of taste. This is how we can refer the taste of one kind of food to another.

Ear

Once the sound or sound wave has entered the drum, it goes to a large structure called the cochlea. In this snail like structure, the sound waves are divided into pitches. The vibrations of the pitches in the cochlea are measured by the Corti. This organ transmits the vibration information to a nerve, which sends it to the brain for interpretation and memory.

Modelling the Microcircuit

The scheme shows the minimal essential building blocks required to reconstruct a neural microcircuit. Microcircuits are composed of neurons and synaptic connections. To model neurons, the three-dimensional morphology, ion channel composition, and distributions and electrical properties of the different types of neuron are required, as well as the total numbers of neurons in the microcircuit and the relative proportions of the different types of neuron. To model synaptic connections, the physiological and pharmacological properties of the different types of synapse that connect any two types of neuron are required, in addition to statistics on which part of the axonal arborization is used (presynaptic innervation pattern) to contact which regions of the target neuron (postsynaptic innervations pattern), how many synapses are involved in forming connections, and the connectivity statistics between any two types of neuron. Neurons receive inputs from thousands of other neurons, which are intricately mapped onto different branches of highly complex dendritic trees and require tens of thousands of compartments to accurately represent them. There is therefore a minimal size of a microcircuit and a minimal complexity of a neuron's morphology that can fully sustain a neuron.



CHAPTER 4

REFLECTION NOTES ON RESULTS

Running the Blue Brain simulation generates huge amounts of data. Analyses of individual neurons must be repeated thousands of times. And analyses dealing with the network activity must deal with data that easily reaches hundreds of gigabytes per second of simulation. Using massively parallel computers the data can be analyzed where it is created (server-side analysis for experimental data, online analysis during simulation). Given the geometric complexity of the column, a visual exploration of the circuit is an important part of the analysis. Mapping the simulation data onto the morphology is invaluable for an immediate verification of single cell activity as well as network phenomena. Architects at EPFL have worked with the Blue Brain developers to design a visualization interface that translates the Blue Gene data into a 3D visual representation of the column. A different supercomputer is used for this computationally intensive task. The visualization of the neurons' shapes is a challenging task given the fact that a column of 10,000 neurons rendered in high quality mesh accounts for essentially 1 billion triangles for which about 100GB of management data is required. Simulation data with a resolution of electrical compartments for each neuron accounts for another 150GB. As the electrical impulse travels through the column, neurons light up and change color as they become electrically active. A visual interface makes it possible to quickly identify areas of interest that can then be studied more extensively using further simulations. A visual representation can also be used to compare the simulation results with experiments that show electrical activity in the brain

4.1 Data Manipulation Cascade

Building the Blue Column requires a series of data manipulations .The first step is to parse each three-dimensional morphology and correct errors due to the in vitro preparation and reconstruction. The repaired neurons are placed in a database from which statistics for the different anatomical classes of neurons are obtained. These statistics are used to clone an indefinite number of neurons in each class to capture the full morphological diversity. The next step is to take each neuron and insert ion channel models in order to produce the array of electrical types. The field has reached a sufficient stage of convergence to generate efforts to classify neurons, such as the Petilla Convention - a conference held in October 2005 on anatomical

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and electrical types of neocortical interneuron, established by the community.

Single-cell gene expression studies of neocortical interneurons now provide detailed predictions of the specific combinations of more than 20 ion channel genes that underlie electrical diversity.

A database of biologically accurate Hodgkin-Huxley ion channel models is being produced.

The simulator NEURON is used with automated fitting algorithms running on Blue Gene to insert ion channels and adjust their parameters to capture the specific electrical properties of the different electrical types found in each anatomical class. The statistical variations within each electrical class are also used to generate subtle variations in discharge behaviour in each neuron. So, each neuron is morphologically and electrically unique. Rather than taking 10,000 days to fit each neuron's electrical behaviour with a unique profile, density and distribution of ion channels, applications are being prepared to use Blue Gene to carry out such a fit in a day. These functionalized neurons are stored in a database. The three-dimensional neurons are then imported into Blue Builder, a circuit builder that loads neurons into their layers according to a "recipe" of neuron numbers and proportions. A collision detection algorithm is run to determine the structural positioning of all axo-dendritic touches, and neurons are jittered and spun until the structural touches match experimentally derived statistics. Probabilities of connectivity between different types of neuron are used to determine which neurons are connected, and all axo-dendritic touches are converted into synaptic connections.

4.2 Whole Brain Simulations

The main limitations for digital computers in the simulation of biological processes are the extreme temporal and spatial resolution demanded by some biological processes, and the limitations of the algorithms that are used to model biological processes. If each atomic collision is simulated, the most powerful supercomputers still take days to simulate a microsecond of protein folding, so it is, of course, not possible to simulate complex biological systems at the atomic scale. However, models at higher levels, such as the molecular or cellular levels, can capture lower-level processes and allow complex large-scale simulations of biological processes. The Blue Brain Project's Blue Gene can simulate a NCC of up to 100,000 highly complex neurons at the cellular or as many as 100 million simple neurons (about the same number of neurons found in a mouse brain). However, simulating neurons embedded in microcircuits,

microcircuits embedded in brain regions, and brain regions embedded in the whole brain as part of the process of understanding the emergence of complex behaviors of animals is an inevitable progression in understanding brain function and dysfunction, and the question is whether whole-brain simulations are at all possible. Computational power needs to increase about 1-million-fold before we will be able to simulate the human brain, with 100 billion neurons, at the same level of detail as the Blue Column. Algorithmic and simulation efficiency (which ensure that all possible FLOPS are exploited) could reduce this requirement by two to three orders of magnitude. Simulating the NCC could also act as a test-bed to refine algorithms required to simulate brain function, which can be used to produce field programmable gate array (FPGA)-based chips. FPGAs could increase computational speeds by as much as two orders of magnitude.

The FPGAs could, in turn, provide the testing ground for the production of specialized NEURON solver applicationspecific integrated circuits (ASICs) that could further increase computational speed by another one to two orders of magnitude. It could therefore be possible, in principle, to simulate the human brain even with current technology. The computer industry is facing what is known as a discontinuity, with increasing processor speed leading to unacceptably high power consumption and heat production. This is pushing a qualitatively new transition in the types of processor to be used in future computers.

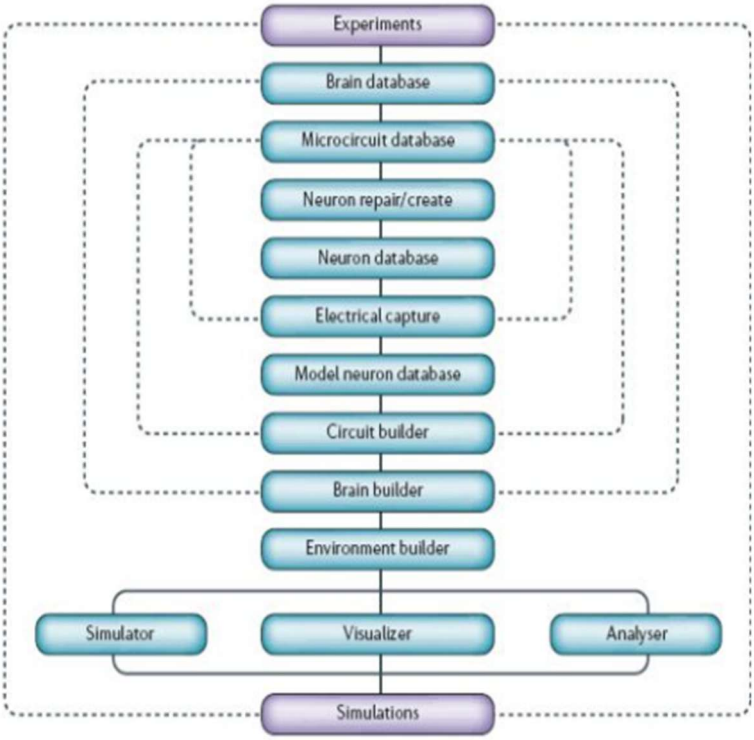


Fig 4.2 The Data Manipulation Cascade

CONCLUSION

In conclusion, we will be able to transfer ourselves into computers at some point. Most arguments against this outcome are seemingly easy to circumvent.

They are either simple minded, or simply require further time for technology to increase. The only serious threats raised are also overcome as we note the combination of biological and digital technologies.

The first version of the Blue Brain, having around ten thousand neurons, has already been created and simulated by some of our great researchers; it is the selection of our deep properties and efficiency of the circuit that takes time. We can aim about deep learning of brain components with its functioning, disfunctioning and impact of a component on another by analyzing different data sets gathered by the past experiences as well as new ones. The amount of time invested to build an appropriate detailed model depends on the depth of the study that is being done and amount of detail that is being captured. The digital era in neurosciences begin with the launch of the human brain project. Fundamentally, there is no problem in modeling a brain and it is therefore likely to be done in the near future. To accomplish this, and hypothesis testing, on our databases can be performed and it is a step into reality.

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